

React Native Package Architecture Specification

Project: Best Buddy (3-Day ADHD Organizer)
Document Purpose: Technical package breakdown mapping directly to client specifications detailed in user sketches and PRD requirements.

1. Summary Technical Stack Overview

To implement the frictionless, encouraging environment requested by the client, the following package dependencies must be integrated into the core React Native repository architecture:

Package Dependency	App Feature Target	Primary Developer Implementation Benefit
react-native-pager-view	3-Day Swipe Navigation	Native hardware-accelerated fluid side-scrolling column navigation.
expo-speech-recognition	"Speak to Text" Entry	Continuous stream dictation avoiding legacy New Architecture crashes.
react-native-mmkv	Automated Midnight Roll-Over	Synchronous super-fast IO checking execution on app state boots.
lottie-react-native	Affirmation & 3-Star Rewards	Vector JSON animations delivering delightful immediate UI feedback.

Package Dependency	App Feature Target	Primary Developer Implementation Benefit
@shopify/flash-list	Categorized Task Groupings	Recycled component views reducing lagging over extensive lists.

2. Detailed Architecture Breakdown by Core Feature

A. Navigation & Fluid Layout (The 3-Day Cycle View)

- **react-native-pager-view**
 - *Purpose:* Essential for executing the client's mandatory side-scrolling framework ("Swipe to next day").
 - *Details:* Allocates 3 distinct view panes mapping dynamically to "Today", "Tomorrow", and "The Next Day".
- **react-native-gesture-handler & react-native-reanimated**
 - *Purpose:* Supports smooth interactive transitions.
 - *Details:* Coordinates shifting base visual hues cleanly (Pale Pink to Melon to Banana) natively on the 60fps UI thread.

B. Low-Friction Entry & Automatic Capturing

- **expo-speech-recognition**
 - *Purpose:* Resolves deprecated microphone streaming libraries to build the client's requested voice portal ("Speak to text box").
 - *Details:* Captures live mic feeds locally to assemble immediate text strings inside task creation modals.
- **linkify-it / react-native-linkify**
 - *Purpose:* Fulfills smart extraction request ("Add any incoming phone numbers or dates").
 - *Details:* Leverages regex heuristics parsing incoming voice text dumps to strip numerical phone tags or calendar dates automatically.

C. Automated Time Shifts & Local Persistence

- **react-native-mmkv**
 - *Purpose:* Serves as the central state anchor evaluating past system checks.
 - *Details:* Bypasses async performance roadblocks to immediately verify current device timestamps during initialization cycles.
- **date-fns / dayjs**
 - *Purpose:* Safely maps shifting relative days without formatting inconsistencies.
 - *Details:* Evaluates local time coordinates to accurately execute the "Roll over at midnight" logic across all 4 categories (Work, Home, Pets, Personal).

D. Gamification & Positive Behavioral Feedback

- **lottie-react-native**
 - *Purpose:* Combats task-evasion and ADHD anxiety using micro-rewards ("Small Reward / Affirmation for accomplishing each task").
 - *Details:* Displays interactive reward layers (like 3-star exploding particle clouds or smiling sun graphics) right when an entry check is toggled.

E. Performance-Driven List Structuring

- **@shopify/flash-list**
 - *Purpose:* Prevents frame-drops across complicated layouts.
 - *Details:* Replaces default flatlist scrolling to cleanly subdivide nested layout sections (e.g., Work -> Prep, Pets -> Wash/Walk, Personal -> Food Diary).